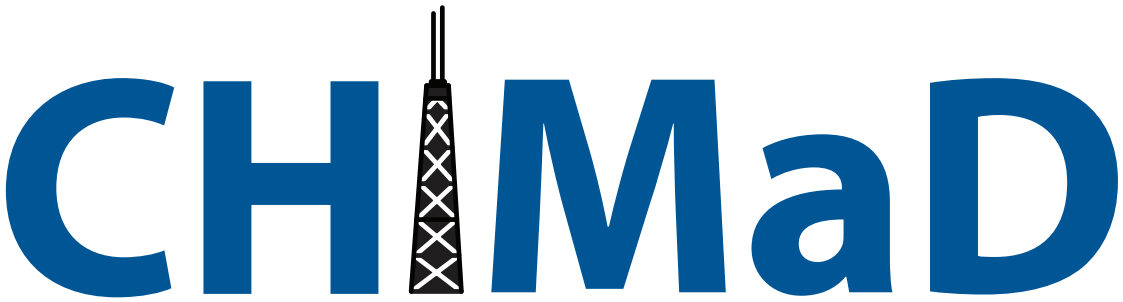


Center for Hierarchical Materials Design

PHASE FIELD METHODS VII

Workshop

NORTHWESTERN UNIVERSITY

CHIMAD.NORTHWESTERN.EDU

September 25-27, 2018

CHiMaD | Center for Hierarchical Materials Design

Center for Hierarchical Materials Design (CHiMaD) is a NIST-sponsored center of excellence for advanced materials research focusing on developing the next generation of computational tools, databases and experimental techniques in order to enable the accelerated design of novel materials and their integration to industry, one of the primary goals of the U.S. administration's Materials Genome Initiative (MGI).

This Chicago-based consortium includes Northwestern University as the lead, University of Chicago, Argonne National Laboratory, and Northwestern-Argonne Institute for Science and Engineering (NAISE), a partnership between Northwestern University. The consortium is also partnered with QuesTek Innovations LLC, a pioneering materials design company, ASM International and Fayetteville State University.

Designing novel materials of specific properties for a particular application requires simultaneously utilizing physical theory, advanced computational methods and models, materials properties databases and complex calculations. This approach stands in contrast to the traditional trial-and-error method of materials discovery. CHiMaD aims to focus this approach on the creation of novel hierarchical materials which exploit distinct structural details at various scales, from the atomic on up, to obtain enhanced properties. The center's research focuses on both organic and inorganic advanced materials in fields as diverse as self-assembled biomaterials, smart materials for self-assembled circuit designs, organic photovoltaic materials, advanced ceramics and metal alloys.

For more information, please visit chimad.northwestern.edu

CHiMaD Contact: E. Begum Gulsoy, e-gulsoy@northwestern.edu

CHiMaD | Center for Hierarchical Materials Design

Agenda
Phase Field Methods VII
September 25-27, 2018

Tuesday, September 25, 2018
CHiMaD Headquarters

12:00PM – 1:00 PM	Registration & Informal Kickoff Lunch
1:00 PM – 1:15 PM	**Welcome & Goals of Workshop** *Olle Heinonen*
1:15 PM – 1:45 PM	**Benchmark problems: Updates** *Olle Heinonen, Andrea Jokisaari*
1:45 PM – 2:15 PM	**PFHub: Updates** *Daniel Wheeler*
2:15 PM – 3:00 PM	**Quantitative Phase field modelling of Continuous Growth Kinetics** *Nikolas Provatas*
3.00 PM	Workshop Group Photo
3:10 PM – 5:00 PM	**Upload of Benchmark Problems** *presentations by students & community on solving the problem; discussion*
5:00 PM – Later	Beer and Dinner on our own

Coffee & snacks available all day

Wednesday, September 26, 2018
CHiMaD Headquarters

8:45 AM	Breakfast
9:00 AM – 9:45 AM	**Teaching using BM problem** *Dong-Uk Kim*
9:45 AM – 10:30 AM	**Predicting precipitate shape morphologies in the presence of coherency stresses and grain-boundary nucleation** *Abhik Choudhury*
10:30 AM – 10:45AM	Break
10:45 AM – 11:30 AM	**Multi-phase-field simulations with OpenPhase** *Marvin Tegeler*
11:30 AM – 12:30 PM	**Phase-field modeling of crystal nucleation I: Fundamentals and methods** *Tamás Pusztai*
12:30 PM – 1:30 PM	Lunch
1:30 PM – 2:30 PM	**Phase-field modeling of crystal nucleation II: Comparison with simulations and experiments** *Laszlo Granasy*
2:30 PM – 3:30 PM	**Using an explicit nucleation model in PRISMS-PF to predict precipitate microstructures** *Steve DeWitt*
3:30 PM – 3:45 PM	Break
3:45 PM – 4:30 PM	**MOOSE-based Nucleation Modeling and Simulation** *Andrea Jokisaari, Larry Aagesen*
4:30 PM – 5:30 PM	Discussion
6:00 PM	Workshop Reception

Coffee & snacks available all day

Thursday, September 27, 2018

8:45 AM	Breakfast
9:00 AM – 10:00 AM	**Phase-field Crystal Modeling of Nucleation Processes** *Laszlo Granasy*
10:00 AM – 10:45 AM	**Machine learning atomistic/molecular models to simulate nucleation and grain growth in polycrystalline materials** *Subramanian Sankaranarayanan*
10:45 AM – 11:00 AM	Break
11:00 AM – 12:00 PM	Wrap-up discussion
12:00 PM	Boxed Lunches to-go & Until next time!

Coffee & snacks available all day

Logistics
Phase Field Methods VII
September 25-27, 2018

Workshop Location

CHiMaD Headquarters
Hogan Building, Suite 1160
Northwestern University
2205 Tech Drive
Evanston, IL 60208

Parking Information

North Campus Parking Garage
2311 North Campus Drive
Evanston, IL, 60208

If driving, please park at the North Garage, and take a ticket as you enter. Validation tickets will be available at CHiMaD Headquarters, see *Victoria Fonseca* to obtain a validation ticket.

Hotel & Shuttle Information

Hilton Orrington
1710 Orrington Ave
Evanston, IL 60201

A shuttle is available to the workshop, but you are also welcome to walk if you prefer. Please see the map on the next page for locations.

Reimbursements

Please contact *Victoria Fonseca*, victoria.fonseca@northwestern.edu for details.